

DAMMU LOHITH KRISHNA BHAGAVAN

Software Engineer · UI/UX & Web Designer · Drone Autonomy · AI/ML · Freelance Developer
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EDUCATION

B.Tech. Computer Science Engineering – GITAM Deemed to be University 2023 – 2026
Diploma in Computer Engineering – Sanketika Polytechnic College 2020 – 2023

TECHNICAL SKILLS

Programming & Web: C, Python, Java, JavaScript (ES6+), HTML5, CSS3, SQL, TCP/IP
AI / ML & Computer Vision: TensorFlow, Keras, PyTorch, OpenCV, Hugging Face, Transformers, Scikit-Learn, YOLO
UI/UX Design: Figma (Wireframing, Prototyping, Auto-Layout, Components, Design Systems, Dev Mode), Adobe XD (UI Flows, Assets, Specs), Canva, InVision, Zeplin
Web Design & Front-End: Responsive Design, CSS Grid, Flexbox, Bootstrap 5, Tailwind CSS, JavaScript animations, WordPress, Wix, Webflow
Embedded & Drone / UAV: Arduino, Raspberry Pi, Pixhawk, ROS, MAVLink, QGroundControl, Mission Planner, RPLIDAR, Jetson Nano, RealSense, Gazebo
Back-End & Tools: Flask, Streamlit, REST APIs, Node.js (npm ecosystem), Git, GitHub, Linux, CUDA, Unity, VS Code, Postman
Cloud & DevOps: AWS (EC2, S3, Lambda – learning)
Frameworks (Developing): React / Angular / Vue (familiarising), E2E Testing (Cypress / Playwright – learning)

EXPERIENCE

Software Engineering Intern | ACCEL UAV Private Limited – acceluav.com May 2025 – Present

- Redesigned acceluav.com end-to-end: stakeholder research → Figma wireframes & high-fidelity prototype → responsive HTML5/CSS3/JS implementation with smooth scroll animations and cross-browser compatibility.
- Built RLTM — a proprietary real-time 2D mapping desktop application for live UAV telemetry visualisation and mission planning using MAVLink data streams.
- Customised QGroundControl (QGC) for VTOL and multirotor fleets; integrated terrain-aligned path planning to reduce collision risk on defense UAV missions.
- Contributed to trajectory planning, homing and guidance algorithms, and system-level integration of mission-ready autonomous UAV platforms.

Freelance UI/UX Designer & Web Developer | Self-Employed 2024 – Present

- Delivered end-to-end web and design projects: client discovery, user research, Figma wireframes + interactive prototypes, and production-ready front-end builds.
- Designed UI systems in Figma — typography scales, colour tokens, reusable component libraries, and handoff specs via Dev Mode / Zeplin for developer teams.
- Built fully responsive, mobile-first websites using HTML5, CSS3, JavaScript, Bootstrap 5, and Tailwind CSS; deployed on WordPress and Webflow for clients.
- Project types: portfolio sites, business landing pages, product showcase pages, e-commerce UIs, SaaS dashboards, and brand identity kits.
- Managed client communication, scoped requirements, set deliverable timelines, and iterated on designs through structured feedback cycles.

Team Lead – Autonomous Aerial Robotics | AIREAL | Center for Autonomous Systems (CAS), GITAM 2024 – 2025

- Led a multidisciplinary team developing AI-powered drones for real-time navigation, obstacle avoidance, and crowd density monitoring.
- Integrated Raspberry Pi, Arduino, ultrasonic sensors, GPS, and ROS into fully autonomous drone navigation pipelines.
- Created team dashboards and internal doc UI in Figma; mentored junior members and managed hardware-software integration milestones.

Machine Learning Intern | HMI Technologies 2022 – 2023

- Developed a Fake Review Detection System in Python using NLP, sentiment analysis, and ML classification; integrated into a Flask web interface.
- Designed the web app UI in Figma and implemented front-end in HTML/CSS/JS for the prototype demonstration.

PROJECTS

- ACCEL UAV Company Website Redesign (UI/UX + Web Dev) · acceluav.com**
Full redesign of acceluav.com for a defense UAV company. Conducted stakeholder interviews and competitive analysis, then produced low-fidelity wireframes and pixel-perfect high-fidelity prototypes in Figma using a custom design system (colour tokens, type scale, icon set, and component library). Implemented the approved design as a fully responsive HTML5/CSS3/JavaScript site with CSS animations, lazy-loading images, and optimised Core Web Vitals for SEO and ATS discoverability.
- Skin Disease Detection Web App (CNN + Flask + Figma UI)**
End-to-end full-stack ML web application for automated skin disease classification. Trained a CNN (TensorFlow/Keras) on a labelled dermatology dataset covering melanoma, eczema, psoriasis, ringworm, acne, and more — achieving 90%+ validation accuracy. Designed the medical-grade UI/UX in Figma (clean card-based layout, accessible colour contrast), then built the front-end in HTML/CSS/JS with a Flask REST API backend for real-time image inference.
- Precision Landing on an ArUco Marker Array (Drone + Computer Vision)**
Precision drone landing system using a downward-facing camera, OpenCV ArUco marker detection, and a PID-based visual servoing loop to correct position and altitude during descent. Fully simulated in Gazebo (ROS integration, physics-accurate environment) before real-world flight tests; achieved consistent sub-10 cm landing accuracy on a static ArUco array.
- AI-Powered Crowd Monitoring Drone – 3rd Place, AI Hackathon 2025**

3-foot H-frame drone with dual Raspberry Pis (4 & 5) and Pixhawk FC; AI-based crowd density analysis pipeline to detect stampede-risk zones and trigger autonomous alerts. Custom Arduino gimbal for stable camera payload. Demonstrated real-time edge computing at Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada (Feb 2025).

- **Face Recognition Attendance System (YOLO + OpenCV)**

YOLO-powered real-time face detection and recognition for automated student attendance logging. OpenCV handles live video capture; recognised faces are matched against a pre-registered database and attendance is logged instantly, eliminating manual errors.

- **Video Call Intercom with Vibration Sensor (SIH 2024 – Accessibility)**

Raspberry Pi hybrid analog/IP intercom system with OpenCV video, IP protocol communication, and vibration-sensor emergency alerts designed for the Deaf and Hard-of-Hearing community. Inclusive UI flows prototyped in Figma before implementation; selected at college level in Smart India Hackathon 2024.

- **Smart Home Automation – ESP32 & Alexa (Sinric Pro)**

Voice-controlled home automation using ESP32 + Sinric Pro + Amazon Alexa. Sensors for temperature, gas, and motion automate safety responses and energy management. Dashboard UI designed in Figma and built with HTML/CSS/JS for local device control.

ACHIEVEMENTS & CERTIFICATIONS

- 3rd Place – AI Hackathon 2025, Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada (Feb 2025)
- Smart India Hackathon 2024 (Internal Edition) – Team Lead; IP-based video intercom for Deaf community accessibility
- Certifications: Python Full-Stack Developer | UI/UX Design Fundamentals | Intro to Programming | SIH 2024 Participation